

MECT2222 Final Exercise Questions

Graphic Solution Notes

How to Use This Sheet

Every solution must start with a free-body diagram. The final exam question file contains vibration, buckling, combined loading, stress transformation, pressure vessel, and shear-flow/shear-stress problems. This document gives graphical solution routes, key formulas, and numeric results where the data are readable.

1 Vibration

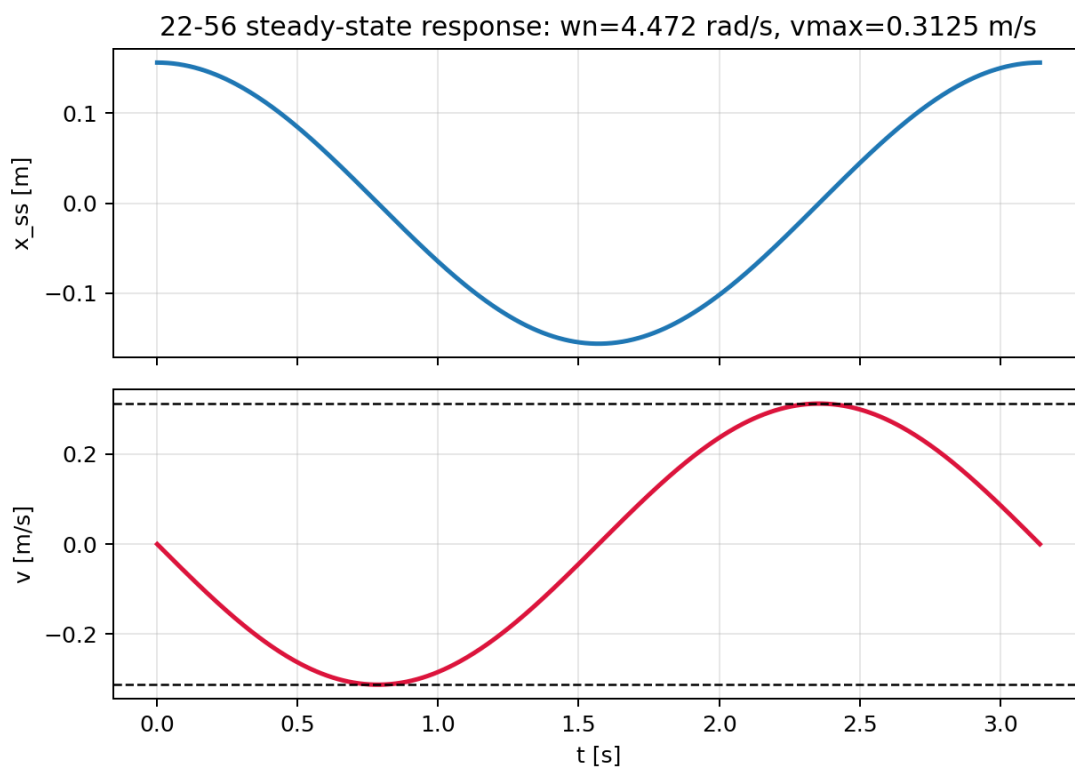
22-56 Forced Undamped Vibration

Given $m = 40 \text{ kg}$, $k = 800 \text{ N/m}$, $F = 100 \cos(2t) \text{ N}$.

$$\omega_n = \sqrt{k/m} = 4.472 \text{ rad/s}$$

$$X = \frac{F_0}{|k - m\omega^2|} = \frac{100}{|800 - 40(2)^2|} = 0.15625 \text{ m}$$

$$v_{\max} = \omega X = 2(0.15625) = 0.3125 \text{ m/s}$$



22-42 General Forced Response

$$mx'' + kx = F_0 \cos \omega t \quad \Rightarrow \quad x'' + \frac{k}{m}x = \frac{F_0}{m} \cos \omega t$$

For $\omega \neq \omega_n$,

$$x = C_1 \cos \omega_n t + C_2 \sin \omega_n t + \frac{F_0}{k - m\omega^2} \cos \omega t$$

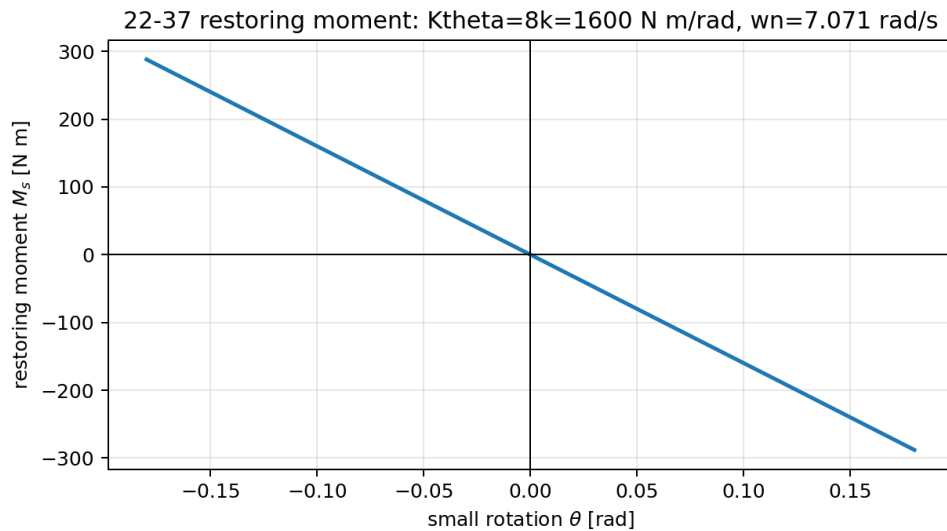
22-37 Slender Rod With Two Springs

For small rotation θ , spring-point displacement is $x = 2\theta$. Restoring moment:

$$M_s = 2k(2\theta)(2) = 8k\theta$$

$$I_O = \frac{1}{3}mL^2 = \frac{1}{3}(6)(4^2) = 32 \text{ kg m}^2$$

$$\omega_n = \sqrt{\frac{8k}{I_O}} = \sqrt{\frac{1600}{32}} = 7.071 \text{ rad/s}$$



22-36, 22-23, 22-15

$$22-36 : \quad k_{eq} = 4k, \quad T_n = 2\pi\sqrt{\frac{m}{4k}}$$

$$22-23 : \quad \omega_n = \sqrt{\frac{4k}{m}} = 5.657 \text{ rad/s}, \quad T_n = 1.111 \text{ s}$$

$$22-15 : \quad \omega_n = \sqrt{\frac{3k}{m}}, \quad T_n = 2\pi\sqrt{\frac{m}{3k}}$$

2 Buckling

13-40

At joint A , member AC is a two-force member with 3-4-5 geometry. For $P = 18 \text{ kN}$,

$$F_{AC} \frac{3}{5} = P, \quad F_{AC} = \frac{5P}{3}$$

Compression in AB :

$$F_{AB} = F_{AC} \frac{4}{5} = \frac{4P}{3} = 24 \text{ kN}$$

Buckling check:

$$P_{cr} = \frac{\pi^2 EI_y}{(KL)^2}, \quad FS = \frac{P_{cr}}{F_{AB}}$$

Use $K = 1$ for pin-pin buckling.

13-34 and 13-13

For frame/deck buckling questions:

$$P_{allow} = \frac{P_{cr}}{FS}, \quad P_{cr} = \frac{\pi^2 EI_{min}}{(KL)^2}$$

Draw the beam/deck FBD first, find column reactions, then compare each reaction with the relevant column buckling load.

3 Shafts and Combined Loading

R11-3, R11-4, 11-43

Use two-plane shaft statics:

$$M_R = \sqrt{M_x^2 + M_z^2}, \quad T = F_t r$$

Solid circular shaft:

$$\sigma = \frac{32M_R}{\pi d^3}, \quad \tau = \frac{16T}{\pi d^3}$$

Von Mises:

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2}$$

Tresca / maximum shear:

$$\tau_{max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

10-87 Pure Torsion, A-36 Steel

For pure torsion with maximum distortion energy:

$$\sigma_{VM} = \sqrt{3}\tau, \quad \tau_{allow} = \frac{S_y}{\sqrt{3}FS}$$

$$\tau = \frac{Tc}{J}, \quad J_s = \frac{\pi d^4}{32}, \quad J_h = \frac{\pi(d_o^4 - d_i^4)}{32}$$

The allowable torque is the smaller value from the solid and hollow segments.

4 Stress Transformation and Failure

10-73 and 10-74

For $\sigma_x = 560 \text{ MPa}$, $\sigma_y = 0$, $\tau_{xy} = 175 \text{ MPa}$:

$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}, \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{1,2} = \sigma_{avg} \pm R$$

Maximum shear theory:

$$S_y \geq \sigma_1 - \sigma_2 = 2R$$

Distortion energy:

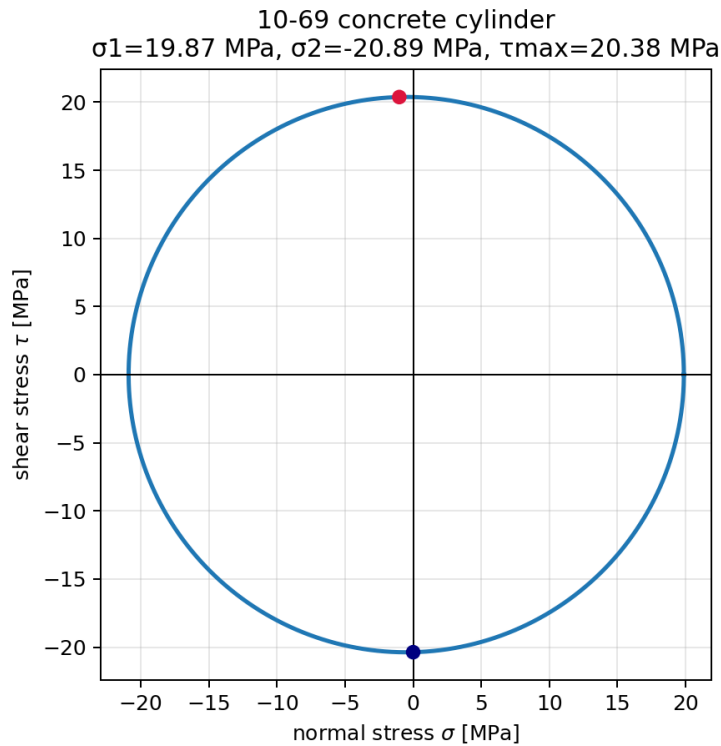
$$S_y \geq \sqrt{\sigma_x^2 - \sigma_x \sigma_y + \sigma_y^2 + 3\tau_{xy}^2}$$

10-69 Concrete Cylinder

$$\sigma = -\frac{P}{A} = -1.019 \text{ MPa}, \quad \tau = \frac{Tc}{J} = 20.372 \text{ MPa}$$

$$\sigma_1 = 19.869 \text{ MPa}, \quad \sigma_2 = -20.887 \text{ MPa}$$

Since $20.887 < 28 \text{ MPa}$, it does not fail by maximum normal stress theory.



10-58 Constrained Material

$$\frac{E_{app}}{E} = \frac{1 - \nu}{(1 + \nu)(1 - 2\nu)}$$

For $\nu = 0.3$:

$$\frac{E_{app}}{E} = \frac{0.7}{1.3(0.4)} = 1.346$$

10-56 and 10-57 Pressure Vessel

$$\sigma_h = \frac{pr}{t}, \quad \sigma_l = \frac{pr}{2t}$$

$$\epsilon_h = \frac{\sigma_h - \nu\sigma_l}{E}, \quad \epsilon_l = \frac{\sigma_l - \nu\sigma_h}{E}$$

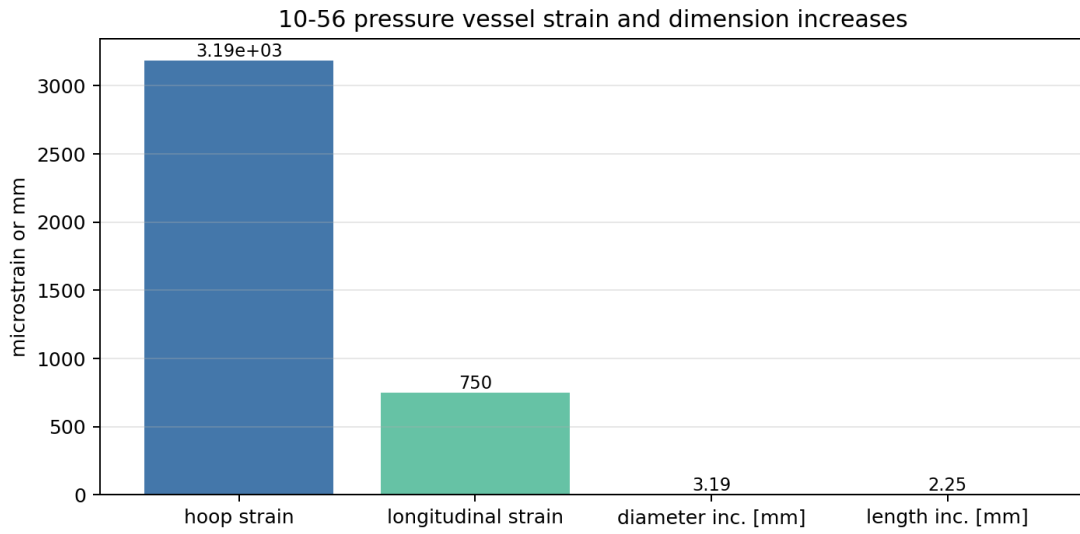
For $p = 15 \text{ MPa}$, $r = 0.5 \text{ m}$, $t = 10 \text{ mm}$, $E = 200 \text{ GPa}$, $\nu = 0.3$:

$$\sigma_h = 750 \text{ MPa}, \quad \sigma_l = 375 \text{ MPa}$$

$$\epsilon_h = 0.0031875, \quad \epsilon_l = 0.000750$$

$$\Delta D = 3.1875 \text{ mm}, \quad \Delta L = 2.25 \text{ mm}$$

$$\frac{\Delta V}{V} \approx 2\epsilon_h + \epsilon_l = 0.007125, \quad \Delta V = 0.0168 \text{ m}^3$$



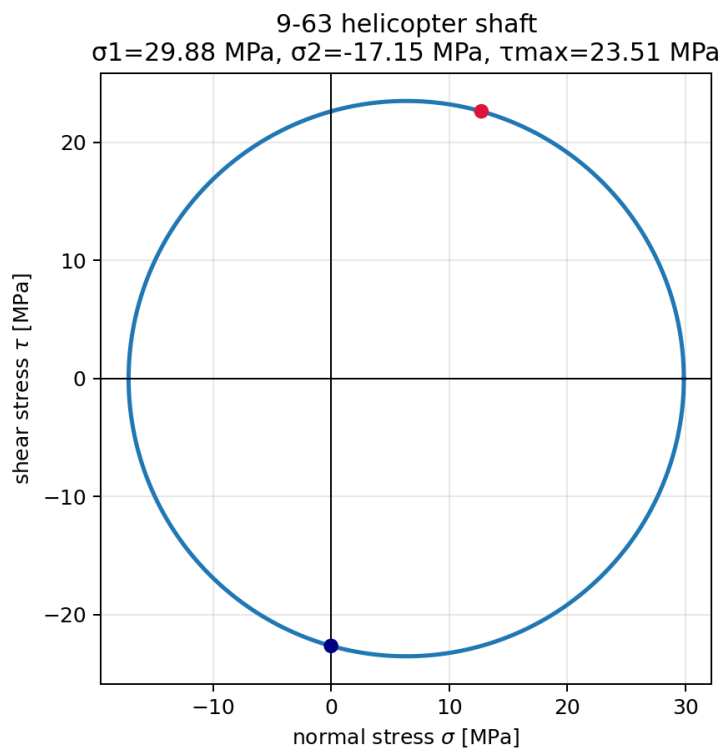
5 Mohr Circle Examples

9-63 Helicopter Rotor Shaft

Given $d = 150 \text{ mm}$, $P = 225 \text{ kN}$, $T = 15 \text{ kNm}$:

$$\sigma = \frac{P}{A} = 12.732 \text{ MPa}, \quad \tau = \frac{Tc}{J} = 22.635 \text{ MPa}$$

$$\sigma_1 = 29.880 \text{ MPa}, \quad \sigma_2 = -17.147 \text{ MPa}, \quad \tau_{\max} = 23.514 \text{ MPa}$$



9-38, 9-39, 9-85, 9-34, 9-35, 9-30

General section stress workflow:

$$\sigma_x = \frac{N}{A} \pm \frac{Mc}{I}$$

$$\tau = \frac{Tc}{J} + \frac{VQ}{It}$$

Then apply:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

6 Shear Stress Problems

7-26

For the simply supported beam:

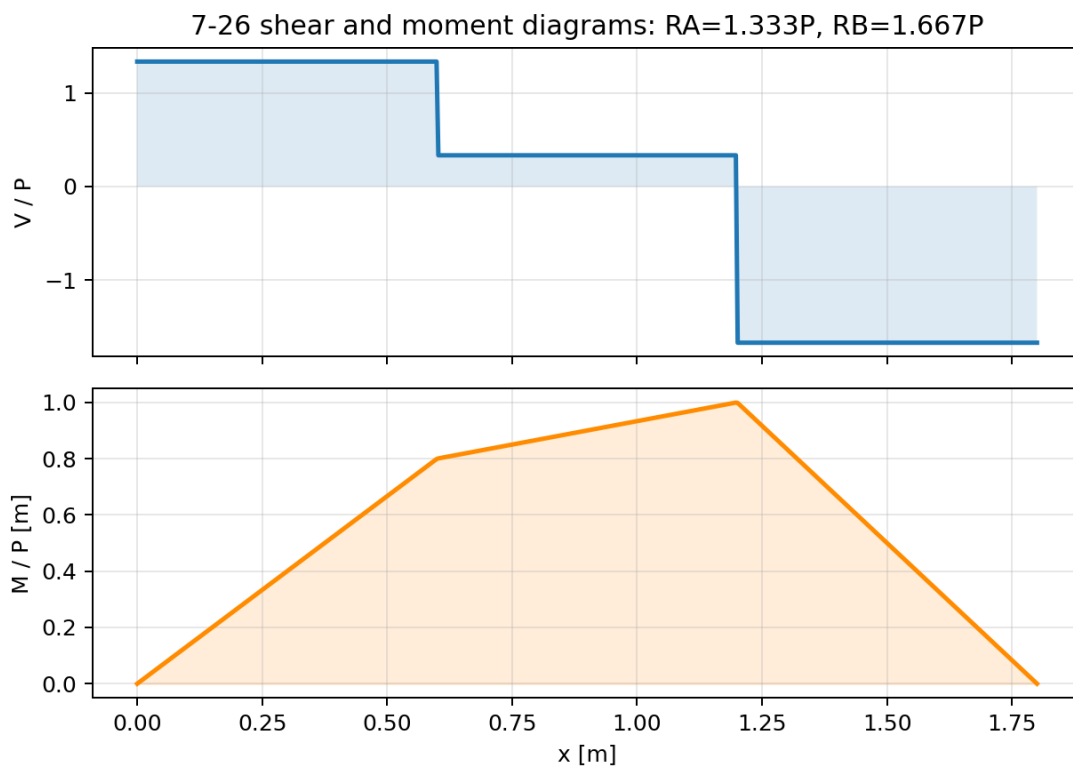
$$R_A = \frac{5P}{3}, \quad R_B = \frac{4P}{3}$$

$$V_{\max} = \frac{5P}{3}$$

Rectangular cross section $b = 50 \text{ mm}$, $d = 100 \text{ mm}$:

$$\tau_{\max} = 1.5 \frac{V}{A}$$

$$3 = 1.5 \frac{(5P/3)}{5000} \quad \Rightarrow \quad P_{\max} = 6000 \text{ N} = 6.0 \text{ kN}$$



7-25

Cantilever rectangular beam:

$$\sigma_{\max} = \frac{PL(h/2)}{bh^3/12} = \frac{6PL}{bh^2}$$
$$\tau_{\max} = 1.5 \frac{P}{bh}$$

Set $\sigma_{\max} = \tau_{\max}$:

$$L = \frac{h}{4}$$

7-22

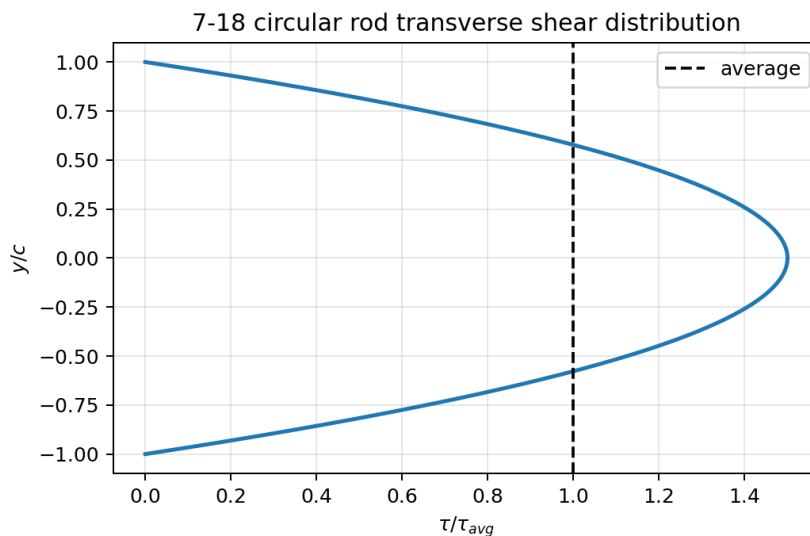
Use:

$$\tau = \frac{VQ}{It}$$

Compute Q at points A and B using the area above or below the point in the web.

7-18 Circular Rod

$$\tau(y) = \frac{4V}{3A} \left(1 - \frac{y^2}{c^2} \right)$$
$$\frac{\tau_{\max}}{\tau_{\text{avg}}} = 4/3$$



Exam Drawing Checklist

1. Draw the global FBD.
2. Draw support reactions and load components.
3. Cut the member at the requested section.
4. Draw N, V, M, T at the cut.
5. Draw stress element at the requested point.
6. Draw Mohr circle if principal stress or maximum shear is requested.
7. Compare with failure criterion when allowable/yield/ultimate stress is given.